Seonji Kim

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Ph.D. Student · KAIST

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About

I am a Ph.D. student in the Graduate School of Culture Technology at KAIST, advised by Prof. Woontack Woo in the Ubiquitous Virtual Reality Lab.

My research focuses on spatial computing and mixed reality, particularly spatial representation, affordance-aware mapping, and adaptive spatial experiences across different physical environments. More recently, I have been exploring how spatial and social contexts shape user experiences, with a growing interest in how AI can support context-aware interactions.

My long-term goal is to develop intelligent spatial computing systems that better understand and adapt to human activities and physical environments. I am currently seeking postdoctoral opportunities to further pursue this direction.

RESEARCH INTERESTS

Spatial Computing · Mixed Reality · Spatial Intelligence

Publications

2026

Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration
Seonji Kim, Dooyoung Kim, Selin Choi, Woontack Woo
IEEE TVCG 32(5), 2026

Task Breakpoint Generation using Origin-Centric Graph in Virtual Reality Recordings for Adaptive Playback
Selin Choi, Dooyoung Kim, Taewook Ha, **Seonji Kim**, Woontack Woo
IEEE TVCG 32(5), 2026

VR Zen Garden: Designing a Virtual Environment for Stress Relief
Hail Song, Jinseok Hong, **Seonji Kim**, Kyung Taek Oh, Woontack Woo, Sungyoung Kim
IEEE VR Workshops, 2026

2025

Architectural Elements Augmentation from 2D Artworks for Spatial Experience of AR Exhibition
Young Bin Kim, Suji Kang, Minju Baeck, **Seonji Kim**, Woontack Woo
IEEE ISMAR Adjunct, 2025

Human-Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph
Taewook Ha, Selin Choi, **Seonji Kim**, Dooyoung Kim, Woontack Woo
IEEE VR Workshops, 2025

Meta-Objects: Interactive and Multisensory Virtual Objects Learned From the Real World for Use in Augmented Reality
Dooyoung Kim, Taewook Ha, Jinseok Hong, **Seonji Kim**, Selin Choi, Heejeong Ko, Woontack Woo
IEEE Computer Graphics and Applications 45(3), 2025

Mutual Space Representation Standardization for Mixed and Augmented Reality Remote Collaboration
Seonji Kim, Woontack Woo
IEEE ISMAR Adjunct, 2025

2024

Experience Graph using Spatio-Temporal Scene Data for Replaying Mixed Reality Interaction
Seonji Kim
IEEE VR Workshops, 2024

Object Cluster Registration of Dissimilar Rooms Using Geometric Spatial Affordance Graph to Generate Shared Virtual Spaces
Seonji Kim, Dooyoung Kim, Jae-eun Shin, Woontack Woo
IEEE VR, 2024

Spatial Affordance-aware Interactable Subspace Allocation for Mixed Reality Telepresence
Dooyoung Kim, **Seonji Kim**, Selin Choi, Woontack Woo
IEEE ISMAR, 2024

2023

The Effects of Spatial Configuration on Relative Translation Gain Thresholds in Redirected Walking
Dooyoung Kim, **Seonji Kim**, Jae-eun Shin, Boram Yoon, Jinwook Kim, Jeongmi Lee, Woontack Woo
Virtual Reality 27(2), 2023

2022

Bring Store in My Room: AR Store Authoring System for Spatial Experience in Mobile Shopping
Seonji Kim, Hyuckjin Jang, Kyung Taek Oh, Seoyoung Oh, Dooyoung Kim, Woontack Woo, Jeongmi Lee, Jaehong Ahn, Sang Ho Yoon
IEEE ISMAR Adjunct, 2022

TranSlider: User-Controlled Augmented Content Presence Visualization in AR Collaboration
Boram Yoon, **Seonji Kim**, Woontack Woo
IEEE ISMAR Adjunct, 2022

Research Experience

2024–Present — **Meta-Object**
Real-time XR Interface Technology Development for Environmental Adaptation
Intelligent Platform Team · IITP, Ministry of Science and ICT

2021–2025 — **OpenXR: TranSpace 3.0**
Development of Untact Realistic OpenXR Platform Technology
Real-World Digital Twin Generation and Management Team

Education

2023–Present — **Ph.D. Student · KAIST**
Graduate School of Culture Technology

2021–2023 — **M.S. · KAIST**
Graduate School of Culture Technology

2016–2021 — **B.S. · Sungkyunkwan University**
Computer Science and Engineering

Teaching

GCT445 — **Augmented Reality**
Teaching Assistant · Summer 2025

GCT600 — **Augmented Reality Project**
Teaching Assistant · Fall 2024

GCT555 — **3D Interaction Design**
Teaching Assistant · Spring 2024

GCT600 — **Augmented Reality Project**
Teaching Assistant · Fall 2023